

Testing past-adapted accounts of anticipatory EEG with post-endpoint randomisation

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Abstract

Anticipatory EEG signatures such as the contingent negative variation are routinely explained by past-adapted, forward-only processes: foreperiod, conditional hazard, arousal, motor preparation and trial history. We ask whether that explanatory class can be tested directly rather than assumed. We describe a design in which a single-trial pre-event neural endpoint is committed and locked before the trial-specific delay to the imperative event is randomly assigned. A delay generated after the endpoint window has closed cannot, under a past-adapted account, order a statistic that was already fixed. The assigned-delay label therefore functions as a negative-control exposure: under valid assignment-to-analysis handling, it should not order a statistic already fixed before assignment. Any material dependence of the committed endpoint on that label localises the problem to randomisation failure, temporal leakage, post-randomisation selection, or residual structure not absorbed by the declared past-adapted analysis in the tested regime. We define a committed pre-event endpoint, a high-capacity forward-only comparator, a participant-level slope estimand, a one-sided materiality threshold, a randomisation test, leakage and delivery audits, and a calibrated omitted-pathway sensitivity analysis. We give the randomisation boundary as a conditional-independence statement and a decision rule with five disjoint outcomes. Across a synthetic benchmark suite the locked pipeline holds its false-positive decision rate at the nominal level under a forward-only data-generating process, recovers an injected residual at the declared materiality boundary at the planned power, and fires the corresponding audits under injected leakage and selection artefacts; realised rates and a fully worked single dataset appear in the electronic supplementary material. These benchmarks establish pipeline validity, not empirical evidence. The contribution is a design-stage falsification test for a widely assumed but rarely tested sufficiency claim about anticipatory EEG.

Keywords: anticipatory EEG; contingent negative variation; temporal expectation; post-endpoint randomisation; negative controls; randomisation inference; design-based falsification.

1 Introduction

The brain prepares for events before they occur. Slow cortical potentials build during a foreperiod, motor systems pre-activate, and pre-stimulus excitability tracks when a stimulus is expected. These

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anticipatory signatures are usually explained by processes that read only the past: elapsed time, cue probability, conditional hazard, arousal, and the recent history of trials. We call such an explanation a *past-adapted account*, because every pre-event output it produces is computed from information available before the event. This view is the default in cognitive neuroscience, and it is well supported.

The default is rarely tested as a claim in its own right. A past-adapted account is a sufficiency statement: it asserts that forward-only structure exhausts the systematic part of pre-event activity. Studies typically confirm that known anticipatory variables predict the endpoint; they less often ask whether anything in the endpoint resists every forward-only explanation. The two questions are different. Confirming that hazard rate predicts the contingent negative variation does not establish that hazard, arousal, and history together leave no residual structure.

This Perspective describes a design that tests the sufficiency claim directly. The idea is a timing boundary. We commit and lock a single-trial pre-event neural endpoint, then randomly assign the trial-specific delay to the imperative event *after* the endpoint window has closed. A delay drawn after the endpoint is fixed cannot have shaped it through any past-adapted process, because it did not exist when the endpoint was formed. The assigned delay therefore acts as a *negative-control exposure*. Under proper randomisation, leakage-safe preprocessing, and an adequate forward-only comparator, the committed endpoint should be conditionally independent of the realised-delay label.

The test is deliberately modest in what it can conclude. A null result would support the adequacy of the forward-only account in the tested regime. A material dependence of the committed endpoint on the realised-delay label would not demonstrate a mechanism, foreknowledge, or any physical interpretation. It would identify one of four ordinary possibilities: a failure of randomisation, a temporal leak in the endpoint or its covariates, a post-randomisation selection pathway through delivery or retention, or conditional insufficiency of the declared forward-only class. Each is checked by a named audit. These four causes of a material association are distinct from the five decision outcomes of Table 1, which additionally cover the null and inconclusive cases and collect the first three causes under a single diagnostic-failure outcome. The design is built so that a surprising result is forced to survive sceptical scrutiny before it is reported as a residual.

The remainder of the article is organised as follows. Section 2 states the forward-only problem and the explanatory class the design must absorb. Section 3 gives the randomisation boundary as a conditional-independence result and its three design conditions. Section 4 defines the committed endpoint, the forward-only comparator, and the participant-level estimand. Section 5 gives the inference, the decision table, the audits, and the omitted-pathway sensitivity analysis. Section 6 reports synthetic pipeline validation and a practical implementation sequence. Section 7 discusses scope, and places any physical interpretation explicitly downstream of and outside the evidential claim of this article.

2 Anticipatory EEG and the forward-only problem

The strongest alternatives to a delay-dependent residual are not generic objections. They are the anticipatory processes already known to shape pre-event EEG. The contingent negative variation is a slow negative potential that develops between a warning cue and an imperative event and scales with temporal expectation¹⁻³. Foreperiod and conditional hazard structure reaction time and preparatory activity, because the probability that an event is imminent rises as time elapses⁴⁻⁷. Attention is allocated in time, and pre-stimulus excitability fluctuates with arousal and ongoing rhythms^{8;9}. Sequential effects carry information from previous trials into the current one¹⁰. Predictive-coding and active-inference

77 accounts frame all of this as the brain using available information to anticipate inputs^{11–15}.

78 These processes share one property that the present design exploits: each is past-adapted. Every
79 quantity they use is measurable before the event. None requires information about a delay that has not
80 yet been generated. A test of sufficiency must therefore give this family its full strength rather than
81 treat it as a set of nuisance regressors added after inspection. We do this by building a high-capacity
82 forward-only comparator from exactly these resources and locking it before any delay labels are seen
83 (Section 4).

84 The design also intersects an older and more contested literature, in which pre-event physiology
85 is reported to track later outcomes^{27;30;31}. That work has been heavily criticised on statistical and
86 replication grounds^{28;29}, and we use it only as a source of design constraints, not as evidence. The
87 lesson we take is procedural: endpoint locking, post-endpoint randomisation, participant-level infer-
88 ence, strictly causal preprocessing, delivery and retention audits, and a prospective selection-sensitivity
89 analysis are the conditions under which any residual would be interpretable, and the conditions whose
90 absence has made earlier claims uninterpretable. Post-endpoint assignment is not itself new: presen-
91 timent paradigms randomise the imperative stimulus after a pre-stimulus physiological window, and
92 the adversarial, consensus-specified Registered Report has applied the same rigour to a claim of that
93 family²⁶. What this design adds is not the timing structure but the inferential apparatus around it:
94 a locked high-capacity forward-only comparator that makes the test one of conditional insufficiency
95 rather than raw association, a graded participant-level slope estimand rather than a binary hit rate,
96 identification by the assignment mechanism with a frozen cross-fitted residual, a calibrated omitted-
97 pathway sensitivity gate, and a non-compensatory decision rule. The contribution is the conversion of
98 a structurally familiar paradigm into a design-based sufficiency test.

99 Methodologically, then, the proposal belongs to the design-based tradition of locked randomisa-
100 tion, covariate adjustment, leakage-safe signal processing, and negative controls, rather than to the
101 evidential tradition of anticipation claims. The assigned delay functions analogously to a negative-
102 control exposure^{21;22}, with one difference worth stating: a classical negative control detects residual
103 confounding in observational data, whereas here randomisation already removes confounding and the
104 delay’s expected null association follows from its generation after the endpoint is fixed. A material
105 association therefore flags not confounding but the named failure modes of randomisation, temporal
106 leakage, post-randomisation selection, or conditional insufficiency.

107 3 Post-endpoint randomisation as a boundary test

108 The force of the design comes from a single choice of ordering: the delay to the imperative event
109 is randomised only after the pre-event endpoint has been committed. This section states what that
110 ordering guarantees and the three conditions it requires.

111 3.1 Endpoint, assignment, and past-adapted information

112 Consider trial j in a warned-foreperiod paradigm with a prospectively fixed endpoint window $I_{\text{pre}} =$
113 $[t_0, t_1]$ and committed closing time t_1 . The single-trial endpoint $A_{\text{pre}}^{(j)}$ is built from a locked channel set,
114 reference interval, temporal operation, aggregation rule, and sign convention, all fixed before access to
115 any delay label, and computed only from data available at or before t_1 .

116 Immediately after the window closes, an independent randomisation service draws the assigned de-
117 lay $\tau_L^{(j)} \in [0, T_0]$ from a prospectively declared scheduler $P_{\text{sch}}(\text{d}\tau_L \mid \mathcal{R}_j)$ within a declared randomisation

118 stratum $\mathcal{R}_j$. Here T_0 is the declared assigned-delay support. The draw is concealed before t_1 from the
 119 participant, acquisition, preprocessing, any adaptive procedure, and the investigators. The apparatus
 120 then attempts delivery after the assigned interval, producing a measured latency $\tilde{\tau}_L^{(j)} = t_{\text{event}}^{(j)} - t_1$ used
 121 for timing-compliance auditing. Under the intention-to-treat principle, the primary analysis is indexed
 122 by the assigned label $\tau_L^{(j)}$, the quantity protected by randomisation, and not by the measured latency.

123 Let $\mathcal{F}_t^{(j)}$ denote the information available on trial j by time $t \leq t_1$: the observed neural and
 124 physiological history, protocol state, elapsed foreperiod, conditional hazard, arousal, motor preparation,
 125 block and session state, time on task, and previous-trial variables. It excludes $\tau_L^{(j)}$, which has not been
 126 generated at endpoint closure. A forward process is *past-adapted* when every pre-assignment output
 127 at time t is $\mathcal{F}_t^{(j)}$ -measurable. This is far broader than a Markov or linear model; it admits nonlinear
 128 state-space, recurrent, hierarchical-latent, predictive-coding, and machine-learning predictors. Because
 129 the stratum label is $\mathcal{F}_{t_1}^{(j)}$ -measurable, the requirement that the service use no pre-assignment input
 130 beyond $\mathcal{R}_j$ implies, under the past-adapted null and an exogenous randomisation source,

$$\tau_L^{(j)} \perp\!\!\!\perp \mathcal{F}_{t_1}^{(j)} \mid \mathcal{R}_j. \quad (1)$$

131 This independence is a statistical implication of the design. It is not a fact read off the assignment
 132 software, and it must be supported by the audits below.

133 3.2 Design conditions (R1)–(R3)

134 The boundary rests on three prospectively audited conditions.

135 **(R1) Post-endpoint randomisation and concealment.** The assigned delay is generated only af-
 136 ter I_{pre} closes, from the declared distribution within $\mathcal{R}_j$, using no trial-specific neural, physiological, be-
 137 havioural, preprocessing, retention, or endpoint input available before t_1 . Under the past-adapted null
 138 the randomisation source is also exogenous to the pre-assignment history, which yields Equation (1).
 139 Logs, timestamps, and seeds are retained, and independent implementation swaps test whether any
 140 association is tied to a particular generator or command pathway.

141 **(R2) Temporal integrity of the endpoint and comparator.** The confirmatory endpoint, ev-
 142 ery confirmatory comparator covariate, and all rules that construct them depend only on samples
 143 available at or before t_1 . Operations that allow post- t_1 samples to change the committed endpoint
 144 are prohibited for confirmatory inference: zero-phase, bidirectional, forward–reverse, symmetric, and
 145 time-shift-corrected filtering are excluded for endpoint construction, as are label-dependent prepro-
 146 cessing, rejection, or inclusion rules. Label-blind artefact models may be used only if their training
 147 data, application window, and rejection rules are fixed before delay access and the temporal-leakage
 148 audit shows that post- t_1 activity cannot alter the committed endpoint beyond the declared tolerance.
 149 The pipeline must pass a prospective temporal-leakage audit in which injected post- t_1 impulses and
 150 step responses produce no committed pre-event output beyond the declared tolerance.

151 **(R3) Assignment-to-analysis integrity.** Delivery, trigger timing, artefact rejection, retention,
 152 missingness, preprocessing, and inclusion must not create a pathway by which the committed endpoint
 153 becomes associated with τ_L . This is not ordinary confounding: post-endpoint randomisation already
 154 protects the assigned delay from pre-assignment causes. It is a post-randomisation selection problem.

An unmeasured pre-event state becomes a threat only if it also affects delivery, retention, censoring, preprocessing, or inclusion in a delay-dependent way. The condition is evaluated by intention-to-treat analysis, delivery and retention audits, negative controls, implementation swaps, and the calibrated selection-sensitivity analysis of Section 5.4.

3.3 What the boundary excludes

Proposition 1 (Conditional exclusion of the past-adapted class). *Let $X^{(j)}$ be an integrable real-valued statistic with randomisation stratum $\mathcal{R}_j$.*

1. Under (R1), if $X^{(j)}$ is $\mathcal{F}_{t_1}^{(j)}$ -measurable, then

$$\mathbb{E}\left[X^{(j)} \mid \sigma\left(\tau_L^{(j)}\right), \mathcal{R}_j\right] = \mathbb{E}\left[X^{(j)} \mid \mathcal{R}_j\right] \quad \text{almost surely.} \quad (2)$$

2. Under (R1)–(R2), Equation (2) applies to the committed endpoint $X^{(j)} = A_{\text{pre}}^{(j)}$ and to any prospectively fixed deterministic transform of pre- t_1 data whose parameters do not depend on the assignment vector, so that the transformed statistic remains $\mathcal{F}_{t_1}^{(j)}$ -measurable.
3. The equality in (2) need not hold within a sample selected after assignment: if an inclusion indicator $S^{(j)}$ depends on both $X^{(j)}$ and $\tau_L^{(j)}$, then conditioning on $S^{(j)} = 1$ can induce dependence between them even when (2) holds unconditionally. The magnitude of any such induced dependence is bounded in SI§8.

The role of Proposition 1 is deliberately narrow. It states the randomisation implication that makes the design discriminating: a statistic fixed before assignment should not acquire an assigned-delay ordering merely because trials are later grouped by τ_L . This is a boundary statement for the proposed experiment. It is not a claim that the comparator has captured every neural cause of anticipation. The proof is elementary and is given in the electronic supplementary material (SI§1). The frozen comparator residual requires a separate design-based argument because the nuisance fit may borrow endpoint information across trials; that argument is given in SI§5.

The retained analysis sample requires an additional qualification. The assigned delay is protected at generation, but the analysed dataset is formed later by delivery, trigger timing, artefact rejection, missingness, preprocessing, and inclusion rules. A valid randomisation record therefore does not by itself validate the analysed contrast. A confirmatory retained-sample claim is made only when the delivery, retention, negative-control, temporal-leakage, and selection-sensitivity checks support that the randomised contrast has been preserved in the analysed sample. If those checks fail, the result is treated as diagnostic or inconclusive rather than as support.

The forward-only comparator has a different role. It improves precision and absorbs known forward-accessible structure, including foreperiod, hazard, arousal, CNV-like activity, sequential history, and time-on-task. The exclusion of an assigned-delay ordering comes from the post-endpoint assignment boundary and the assignment-to-analysis checks, not from asking the comparator to prove that all anticipatory mechanisms have been modelled.

4 Endpoint, estimand and comparator

This section fixes the three objects that must be locked before any delay label is seen: the committed endpoint, the forward-only comparator, and the participant-level estimand.

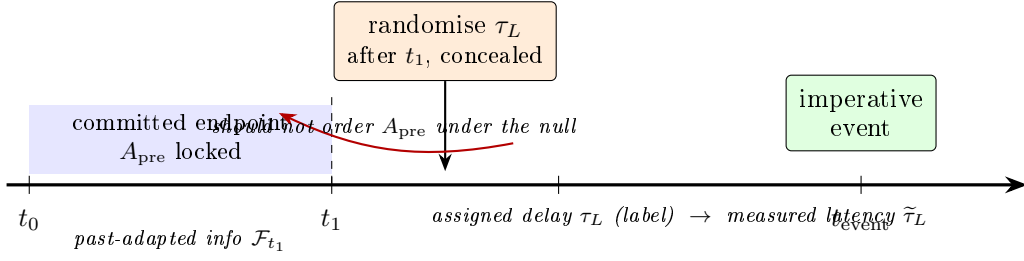


Figure 1: Post-endpoint randomisation boundary. The single-trial endpoint A_{pre} is committed and locked over $I_{\text{pre}} = [t_0, t_1]$ using only past-adapted information $\mathcal{F}_{t_1}$. The trial-specific delay τ_L is randomised only after t_1 and concealed before it. Under (R1)–(R2), the assigned delay τ_L cannot order the committed endpoint A_{pre} at the design stage; under (R3), this exclusion is carried into the retained confirmatory analysis (Proposition 1). The measured latency $\tilde{\tau}_L$ is used for auditing; the primary analysis is indexed by the assigned label τ_L .

4.1 Committed pre-event endpoint

The endpoint $A_{\text{pre}}^{(j)}$ is a prospectively locked, signed, baseline-corrected single-trial feature from a committed pre-event window and electrode set. The locked specification fixes the recording modality, channel or source set, reference, baseline interval, endpoint window, temporal operation, aggregation rule, sign convention, and trial-eligibility criteria. Confirmatory preprocessing is strictly causal: an output at time t may depend only on samples acquired at $u \leq t$. Acausal variants may be reported only as exploratory robustness analyses and cannot enter the confirmatory decision. The endpoint need not be computed online at t_1 , but its construction rule, eligibility rule, and sign convention become irrevocable at endpoint closure, and later computation must remain blind to delay labels.

4.2 Forward-only comparator

The forward-only comparator is the pre-specified high-capacity past-adapted predictor whose job is to absorb known anticipatory structure, reduce residual variance, and test robustness against a strong set of conventional explanations. Its covariate set is broad but not magical. It may include elapsed foreperiod, conditional hazard, scheduler and block-probability information, a CNV-like slow potential amplitude, an arousal or vigilance proxy, motor-preparation proxies, sequential trial history, reaction-time history, time on task, preprocessing covariates, participant and session structure, and prospectively declared interactions and nonlinear transforms. Every input must be measurable with respect to pre-assignment information $\mathcal{F}_{t_1}^{(j)}$. No realised-delay label, and no deterministic transform of a delay label, may enter comparator fitting.

Let X_j denote the locked covariate vector and $m_0(X_j) = \mathbb{E}[A_{\text{pre}}^{(j)} | X_j]$ the comparator target. The family, tuning rule, cross-fitting fold structure, and loss are fixed before delay access. The comparator family may include penalised regression, gradient boosting, random forests, or other pre-specified supervised predictors, provided the family, tuning rule, fold structure, and loss are locked before delay access^{16;17}. Folds respect randomisation blocks, and all hyperparameter selection occurs inside the corresponding training data. For a trial in fold $k(j)$ the confirmatory residual is

$$A_{\text{pre}}^{\text{resid},(j)} = A_{\text{pre}}^{(j)} - \hat{m}_0^{(-k(j))}(X_j). \quad (3)$$

Held-out predictions and residuals are generated once and frozen before the randomisation test; refitting the forward-only comparator inside the test is prohibited. Residualisation improves precision and

probes robustness. It is not the source of identification. The delay contrast is identified by post-endpoint randomisation, following the design-based logic of covariate-adjusted randomisation inference, in which nuisance adjustment improves precision while the contrast is justified by the assignment mechanism rather than by the predictive model^{18;19}.

Concrete covariate constructions

To make the comparator class concrete, we give three constructions that together absorb the dominant anticipatory signatures. The paradigm is assumed to vary the *cue-to-endpoint* foreperiod across trials, so that conventional preparation varies trial to trial and is fully past-adapted, while the separate post-endpoint delay τ_L is the randomised quantity targeted by the test. Each construction below is $\mathcal{F}_{t_1}^{(j)}$ -measurable, uses only the declared, participant-learned schedule rather than the realised draw, and enters X_j directly.

First, a CNV-like slow-potential covariate. Let $V_c^{(j)}(t)$ be a causally low-pass-filtered fronto-central potential, baseline-corrected to the cue-locked baseline, and let $W_{\text{cnv}} = [t_1 - \Delta_{\text{cnv}}, t_1)$ be a pre-endpoint window declared disjoint from the endpoint feature support. The covariate is the ordinary-least-squares slope of the slow potential over that window,

$$x_{\text{cnv}}^{(j)} = \frac{\sum_{t \in W_{\text{cnv}}} (t - \bar{t}) (V_c^{(j)}(t) - \bar{V}_c^{(j)})}{\sum_{t \in W_{\text{cnv}}} (t - \bar{t})^2}, \quad (4)$$

which captures the rate of the expectancy-related negativity that develops before the endpoint closes^{1;3}.

Second, a conditional-hazard covariate. Let G be the declared (learned) cumulative distribution of the cue-to-endpoint foreperiod, g its density, and $e^{(j)} = t_1 - t_{\text{cue}}^{(j)}$ the elapsed foreperiod at endpoint closure. The instantaneous and integrated hazard covariates are

$$\lambda^{(j)} = \frac{g(e^{(j)})}{1 - G(e^{(j)})}, \quad \Lambda^{(j)} = -\log(1 - G(e^{(j)})), \quad (5)$$

evaluated from the declared distribution at the elapsed foreperiod. They encode the rising conditional probability that the event is imminent and use no realised τ_L ^{4;6;7}.

Third, a sequential foreperiod-history covariate. Preparation on trial j depends on the previous trial's timing, the classical sequential effect¹⁰. We include the previous elapsed foreperiod and its signed change,

$$x_{\text{seq}}^{(j)} = e^{(j-1)}, \quad \Delta e^{(j)} = e^{(j)} - e^{(j-1)}, \quad (6)$$

optionally with the previous-trial response or outcome. All three constructions, their windows, filters, and any interactions are locked before delay access and are computed by the same strictly causal, label-blind pipeline as the endpoint. One subtlety deserves note: because the participant cannot perceive the endpoint closure, their anticipatory state at t_1 reflects the distribution of the full cue-to-event interval, which the cue-to-endpoint hazard of (5) only approximates. This misspecification affects only the residual variance, and hence power, not the conditional-independence guarantee of Proposition 1, which holds for any $\mathcal{F}_{t_1}^{(j)}$ -measurable comparator regardless of its accuracy. A hazard built from the full-interval distribution, obtained by convolving the foreperiod schedule with the marginal delay law, may be added as a locked covariate to recover precision.

4.3 Participant-level estimand

The participant is the independent inferential unit; trials from one participant are not independent biological replications. Within participant p , trial j , and stratum $\mathcal{R}_{pj}$, the assigned delay is centred at its known scheduler mean and the participant slope is defined by

$$A_{\text{pre}}^{\text{resid},(pj)} = \alpha_{\mathcal{R}_{pj}} + \beta_{\tau,p} \left\{ \tau_L^{(pj)} - \mathbb{E}_{P_{\text{sch}}}[\tau_L \mid \mathcal{R}_{pj}] \right\} + \varepsilon_{pj}. \quad (7)$$

The population estimand is the equal-participant average

$$\beta_{\tau} = \mathbb{E}_p[\beta_{\tau,p}]. \quad (8)$$

The coefficient β_{τ} is a signed linear summary over the declared delay support. It assumes no exponential shape and no extrapolation beyond that support. Because the endpoint is committed before assignment, β_{τ} is not a forward-time treatment effect of delay on an earlier endpoint; it is the slope of a negative-control exposure. Two inferential routes are declared prospectively. The assignment-isolation route holds the frozen endpoint array fixed under reassignment and is used only when reset, washout, and lagged-delay diagnostics support endpoint-array invariance. When the assigned delay on trial j may affect later endpoints through carryover, adaptation, fatigue, or sequential history, the analysis uses the sequential-score route in SI §4, which treats each assignment as a conditionally random draw given the observed pre-assignment history.

5 Decision rule, audits and sensitivity analysis

The confirmatory decision combines a randomisation test, a participant-level materiality bound, a set of audits, and a selection-sensitivity bound. The commitments below are fixed before any delay label is analysed.

5.1 Hypotheses, materiality and inference

The predeclared directional hypotheses are

$$H_0 : \beta_{\tau} \geq 0, \quad H_1 : \beta_{\tau} < 0. \quad (9)$$

The one-sided sign is a substantive, pre-registered prior, not a property of the design: we test specifically for a residual largest at the shortest post-endpoint delays and decaying as the delay grows, the form a genuine post-endpoint dependence would be expected to take. The design itself is sign-symmetric. Under (R1)–(R3) and the past-adapted null the assigned delay should show no material association of either sign, and a material association of either sign that survives every audit is, by Proposition 1, evidence that the declared forward-only account is conditionally insufficient in the tested regime. A surviving negative slope additionally confirms the pre-registered prediction; a surviving positive slope is conditional insufficiency in the unpredicted direction. The sign decides whether the directional prediction is confirmed, not whether the result is real: that is decided by the sign-blind audits, equally for both directions.

Let $\beta_{\min} \geq 0$ be the prospectively declared minimum material magnitude, in the endpoint's native

units per second or on a fixed label-blind harmonised scale. The materiality criterion is

$$\text{UCB}_{0.95}(\hat{\beta}_\tau) < -\beta_{\min}. \quad (10)$$

Because no validated effect-size scale exists for a delay residual, $\beta_{\min}$ is a single-participant resolution floor rather than a magnitude-of-importance threshold or a fixed population significance level. It is the label-blind resolvability of one participant's slope,

$$\beta_{\min} = \kappa \frac{\sigma_{\text{resid}}^{\text{blind}}}{\sigma_\tau \sqrt{\bar{n}_{\text{ret}}}}, \quad (11)$$

with $\sigma_{\text{resid}}^{\text{blind}}$ the label-blind within-participant residual scale, σ_τ the assigned-delay standard deviation over the declared support, $\bar{n}_{\text{ret}}$ the retained trials per participant, and κ a prospectively fixed multiple. The floor is not a fixed α -level test: the final support decision combines the randomisation test, the participant-level upper bound, this resolution floor, the audits, $N_{\min}$, and the selection gate. The effective normal-equivalent stringency of the upper-bound component depends on $\beta_{\min}/\text{se}_{\text{pop}}$; under a normal null approximation $\Pr\{\text{UCB}_{0.95}(\hat{\beta}_\tau) < -\beta_{\min}\} \approx \Phi[-(\beta_{\min}/\text{se}_{\text{pop}} + 1.645)]$, but this approximation varies by many orders of magnitude with se_{pop} and is not a declared test size. The confirmatory calibration is empirical and design-based, using the randomisation distribution and simulation, not this approximation. The floor tightens as the sample grows and guards against false positives, not against small-but-resolvable effects. The unstandardised slope and its physical interpretation are always reported so that scientific importance can be judged directly. It is fixed by a registered formula evaluated only on label-blind qualification data, never from the observed $\hat{\beta}_\tau$, an injected synthetic effect, or an unblinded pilot.

Inference uses two procedures with distinct roles. The assignment-calibrated test conditions on the committed endpoints, frozen residuals, and participant and stratum structure, and asks whether the observed statistic is unusually negative under the assignment mechanism actually used. Each replicate permutes fixed multisets within declared blocks or redraws stochastic assignments from P_{sch} ; no replicate moves an assignment across participant, session, or stratum. The one-sided randomisation p -value is the probability of a statistic at least as negative as observed, with a finite-sample plus-one correction under Monte Carlo enumeration²⁰. The participant-level upper confidence bound $\text{UCB}_{0.95}(\hat{\beta}_\tau)$ comes from a studentised participant bootstrap- t , resampling participants with all trials and frozen residuals intact; a bias-corrected and accelerated bootstrap and a participant t -interval are reported as sensitivity analyses. A minimum analysable participant count $N_{\min}$ is fixed by simulation under the planned operating characteristics; below it, the assignment-calibrated result may be reported but no population-materiality conclusion is drawn. Estimator details are given in SI §6.

5.2 Decision rule

A dataset receives support only when all of the following hold:

1. the one-sided assignment-calibrated criterion passes;
2. $\text{UCB}_{0.95}(\hat{\beta}_\tau) < -\beta_{\min}$;
3. the analysable participant count is at least $N_{\min}$;
4. the randomisation, temporal-leakage, delivery, retention, and applicable implementation-swap audits pass;

Table 1: Estimand, null and decision outcomes for the post-endpoint randomisation test. The estimand is the population participant-level slope β_τ of Equation (8); the null is $\beta_\tau \geq 0$.

Outcome	Statistical pattern	Interpretation
Supported residual	Randomisation test rejects; $UCB_{0.95}(\hat{\beta}_\tau) < -\beta_{\min}$; $N \geq N_{\min}$; audits and sensitivity bound pass.	A materially negative assigned-delay ordering remains after the declared past-adapted analysis, conditional on the randomisation, leakage, delivery, retention, and selection-sensitivity checks. No mechanism is implied; programme-level support requires independent replication.
Forward-only adequate (null)	No material negative residual; audits pass; design adequately sensitive.	Within the declared sensitivity envelope, the forward-only account is adequate for the tested endpoint, regime, and delay support.
Diagnostic failure	Randomisation imbalance, temporal-leakage detection, preprocessing dependence, or a failed negative control.	The test is invalid or uninterpretable. Resolve by redesign and new data; not evidence about sufficiency.
Opposite-direction departure	Material positive slope; audits and sensitivity scrutiny applied.	Incompatible with the pre-registered directional prediction. If it survives every audit and the selection gate, it is conditional insufficiency of the past-adapted account in the unpredicted direction, reported as such; it does not confirm the prediction and requires replication.
Ambiguous / inconclusive	Randomisation test and materiality bound disagree, or $N < N_{\min}$.	Reported descriptively; assignment-calibrated or population-materiality evidence is insufficient for a confirmatory conclusion.

5. the prospectively calibrated selection-sensitivity bound passes (Section 5.4);

6. the prospectively declared endpoint-by-delay collider diagnostic does not fire (Section 5.4). This is a separate requirement from the scalar selection-sensitivity bound, which is built on the marginal retention imbalance and cannot detect a balanced collider.

These form one non-compensatory intersection rule; the components are not interchangeable, and no post-result rule selects whichever procedure is more favourable. Table 1 lists the resulting outcomes. Programme-level support additionally requires independent replication using the same locked endpoint, comparator family, materiality threshold, inference, and audit logic. A single positive dataset supports only that dataset.

5.3 Audits and negative controls

Four audits protect the three conditions and are reported regardless of outcome. The randomisation audit checks assignment balance across strata, seed and log integrity, and concealment (R1). The temporal-leakage audit injects post-event impulses and step responses into the locked pipeline and requires no committed pre-event output beyond tolerance (R2). The delivery audit compares assigned labels with measured latencies and bounds timing non-compliance. The retention audit records stratum-specific delivery, rejection, missingness, and inclusion, including the retained-against-excluded endpoint distributions available before unblinding (R3).

Negative controls are two-sided diagnostics. A negative-control exposure or outcome that should be unrelated to the contrast, but shows a material association in either direction, indicates that randomisation, leakage, or no-selection assumptions require investigation^{21;22}. The assigned delay is itself the primary negative-control exposure; only a materially *negative* slope is the proposed residual, while a material association of either sign on an auxiliary negative control is a design alarm. Audit failure is classified separately from a null: a failed audit makes the test invalid, not evidence that the past-adapted account is sufficient.

5.4 Omitted-pathway sensitivity analysis

Condition (R3) cannot be closed by inspection, so its residual risk is quantified. The selection-sensitivity analysis reports the smallest differential, endpoint-correlated retention or inclusion imbalance across the delay support that could reproduce the observed slope under an otherwise past-adapted account, and compares it with the imbalance bounded by the audits. The mapping from an imbalance to an induced slope is computed under a declared selection-model class, using monotone sharp bounds²³ and worst-case bounds without monotonicity²⁴, in the sensitivity-analysis tradition for observational threats²⁵. The gate compares confidence limits, not point estimates: support requires the lower confidence limit on the imbalance needed to manufacture the slope to exceed the upper confidence limit on the imbalance the audits permit. If the slope is material but this gate is not cleared, the result is reported as selection-limited, not as support. Full construction is given in SI §8.

The scalar selection-sensitivity gate protects only against selection pathways represented by the declared audited imbalance summary and selection-model class. It is built on the *marginal* retention imbalance across delay bins, and marginal retention balance is not sufficient to exclude selection. A pure endpoint-by-delay collider, in which inclusion depends on the joint configuration of the committed endpoint and the assigned delay with no main delay effect on retention, can manufacture a retained-sample slope while marginal retention rates by delay bin remain approximately balanced; the full pre-selection sample still satisfies the post-endpoint randomisation boundary, and the association arises only after conditioning on the retained analysis sample. Because the committed endpoint exists for every trial, this pathway is detectable directly: a prospectively declared endpoint-by-delay diagnostic tests whether inclusion depends on the endpoint-by-delay interaction, and the retained-against-excluded endpoint distributions are compared within delay bins. If this diagnostic fires, or if the retained-sample slope is compatible with an uncovered collider of the observed magnitude under approximately balanced marginal retention, the dataset cannot receive supported-residual classification and is reported as selection-limited or operationally unqualified. In the benchmark this diagnostic, not the scalar gate, is what blocks a manufactured collider slope (Section 6.1; SI §8–§9).

6 Synthetic validation and practical implementation

6.1 What the synthetic benchmarks establish

The benchmarks are the pre-registered qualification of the analysis pipeline, and their realised operating characteristics are reported in SI §9 and were produced by the public benchmark package (Section 7.1). They do not provide evidence that a delay residual exists in human EEG. Before any delay labels are analysed, the locked pipeline is run on simulated data with known generating processes, and is required to behave as designed across seven scenarios. Under a pure forward-only data-generating process, the delay slope must be null after the comparator: the randomisation test should reject at its nominal rate and the materiality bound should not clear $-\beta_{\min}$. Under an endpoint-level injected residual, the pipeline must recover the negative sign and approximate magnitude with the planned estimators and sample sizes. Under an injected leakage or retention artefact, the audits must detect it. Under a standard selection model the sensitivity bound must move as the marginal imbalance is varied, and under a pure endpoint-by-delay collider the dataset must be classified selection-limited rather than supported. Under an adversarial forward-only null with nonlinear hazard structure, heavy-tailed participant heterogeneity, heteroskedastic and autocorrelated residuals, carryover, and comparator

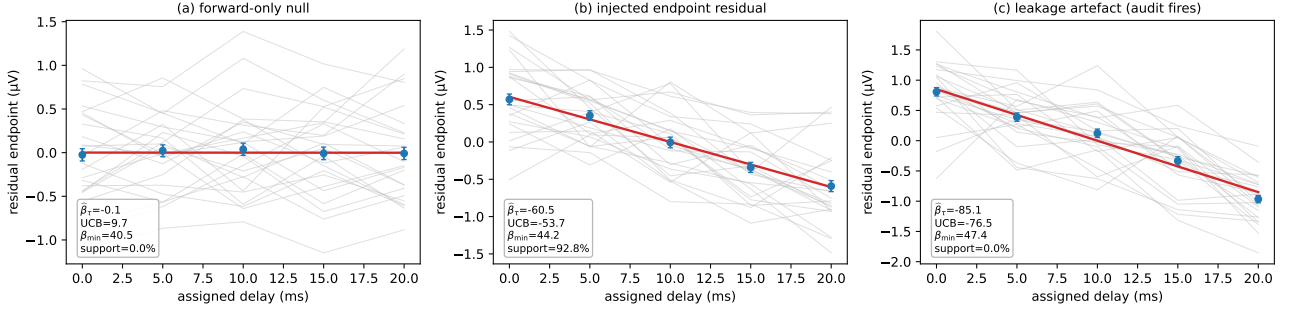


Figure 2: Synthetic validation of the Level II-A decision pipeline using simulated data, not human EEG. Anchor grid $T_0 = 20$ ms with five assigned-delay bins $\{0, 5, 10, 15, 20\}$ ms, 24 participants, and residual noise scale $1 \mu V$. Within each scenario the displayed dataset is the Monte Carlo replicate whose $\hat{\beta}_\tau$ is closest to the scenario median among replicates with the modal final decision. Faint lines are participant-level bin means, filled markers are grand bin means $\pm SE$, and the red line is the fitted population slope. Annotations report $\hat{\beta}_\tau$, the one-sided upper bound $UCB_{0.95}$, the resolution floor $\beta_{\min}$ for the scenario, and the scenario support rate. Operating rates summarise $M = 1000$ simulated datasets per generator (run hash 0a775ac4f1ead9d3). (a) Under a forward-only generator the residual carries no assigned-delay slope; false support is obtained in 0/1000 datasets, with the randomisation test rejecting at its nominal 5% rate. (b) With an endpoint-level injected slope of $-60 \mu V s^{-1}$ added to the committed endpoint before comparator fitting, the pipeline recovers the negative residual and supports it in 92.8% of datasets. (c) A post- t_1 temporal leak manufactures an apparent slope that clears the materiality floor, but the temporal-leakage audit exceeds tolerance and fires in all 1000 datasets, so the non-compensatory rule blocks support. Selection artefacts, including a balanced endpoint-by-delay collider, are handled separately by the selection-sensitivity gate and the endpoint-by-delay diagnostic (SI §8–§9).

misspecification, false-support control must hold. Under an opposite-direction injection the result must be classified as an opposite-direction departure. A pipeline that fails any of these is reported as operationally unqualified rather than as evidence for or against the sufficiency claim.

In the realised run (run hash 0a775ac4f1ead9d3, $M = 1000$ datasets per scenario), the forward-only anchor produced support in 0/1000 datasets with the randomisation test rejecting at 0.050; the adversarial forward-only null also produced 0/1000 support. The endpoint-level injected residual of $-60 \mu V s^{-1}$ was supported in 92.8% of datasets. The leakage artefact fired the temporal-leakage audit in all datasets and was never supported. The standard selection artefact fired the retention audit in all datasets and was never supported. The pure endpoint-by-delay collider, calibrated to approximately balanced marginal retention, manufactured a material retained-sample slope that the scalar selection gate passed in 98% of datasets, yet the endpoint-by-delay diagnostic fired in all datasets, so it was classified selection-limited in 97.9% and supported in 0%. The opposite-direction injection was classified as an opposite-direction departure in 95.3% and never as directional support. Full realised operating characteristics, including the collider scope sweep, are tabulated in SI §9.

6.2 Synthetic anchor and planning power

The reference benchmark uses a deliberately short synthetic support $T_0 = 20$ ms with five assigned-delay bins $\{0, 5, 10, 15, 20\}$ ms and, where a residual is injected, an additive slope at the endpoint-generating level. This grid is not proposed as the empirical horizon for anticipatory EEG. It is a low-leverage stress test of the estimator: ordinary long-foreperiod structure is kept small, and any

recovered slope must come from the injected residual or from a deliberately injected artefact.

The injected residual is added to the committed endpoint before comparator fitting and residualisation, not to the post-comparator residual. When the past-adapted covariates are independent of the assigned delay by construction and the comparator is label-blind, endpoint-level and residual-scale injection are close in the clean anchor. The endpoint-level benchmark is nevertheless the confirmatory benchmark because it exercises the comparator, cross-fitting, freezing, residualisation, randomisation test, bootstrap bound, and decision rule as a single coupled pipeline.

In the retained-sample synthetic benchmark, the estimator reaches the planning target only when the injected slope is large relative to the residual noise and the retained bins are balanced. The favourable grid reaches the target with about five retained trials per bin, whereas the higher-noise grid requires about twenty retained trials per bin. These values are synthetic estimator operating characteristics, not EEG sample-size recommendations. An empirical protocol must translate them into a participant-by-trial design using label-blind estimates of residual variance, participant heterogeneity, retained-trial yield, artefact loss, timing error, delivery compliance, and the declared materiality threshold.

The admissible delay support is itself a design variable with opposing effects: a wider support increases assigned-delay variance and statistical leverage, but also lengthens the interval over which ordinary foreperiod and hazard structure operate, enlarging the burden on the comparator. The support is fixed in a label-blind qualification sample at the point that balances leverage against the forward-accessible structure the comparator must absorb, and is not revised using delay information. Benchmark specifications and the planning grid are given in SI §9.

6.3 Practical implementation sequence

The most economical programme is staged. First, qualify the hardware, scheduler, endpoint, concealment, delivery logging, retention rules, and analysis implementation using synthetic signals, hardware loopback data, and an independent non-confirmatory pilot. Second, convert the design into a full empirical protocol, preferably for Stage 1 Registered Report review, fixing the endpoint, scheduler, comparator, materiality threshold, sample-size rule, exclusions, audits, selection-sensitivity analysis, and decision rule at the level required for prospective execution. Third, run the dataset-level test and report the audit and feasibility record regardless of outcome. Fourth, repeat the same locked test in an independent sample. This sequence minimises analytic flexibility and makes a surprising result expensive to fake and easy to falsify. Figure 3 summarises the analysis flow and the leakage barriers that separate label-blind locking from confirmatory inference.

7 Discussion

This article describes a design that converts a standing assumption of anticipatory neuroscience into a testable boundary. The assumption is that past-adapted, forward-only processes exhaust the systematic part of pre-event EEG. The boundary is a timing choice: commit the endpoint, then randomise the delay. Under proper randomisation, leakage-safe preprocessing, and an adequate forward-only comparator, the committed endpoint is conditionally independent of the realised-delay label (Proposition 1). The estimand, materiality threshold, decision rule, audits, and sensitivity analysis make the test falsifiable in advance and interpretable after the fact.

The warranted conclusions are deliberately narrow. A valid, adequately sensitive null supports

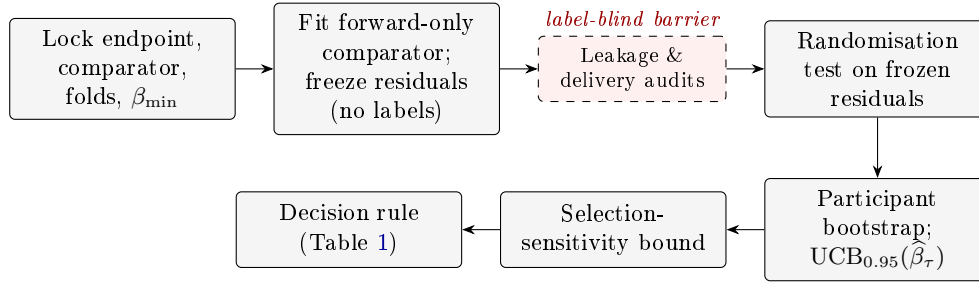


Figure 3: Analysis pipeline and leakage barriers. Endpoint, comparator, folds, and materiality threshold are locked, and the forward-only comparator residuals are frozen, before any delay label is accessed. Leakage and delivery audits guard the barrier between label-blind construction and confirmatory inference. The randomisation test, participant bootstrap, and selection-sensitivity bound feed the non-compensatory decision rule.

the forward-only account within the declared sensitivity envelope for the tested endpoint, regime, and delay support; it does not prove any forward model universally sufficient. A supported residual establishes conditional insufficiency of the declared past-adapted class in that regime, or a failure of a design assumption that the audits did not catch. It does not, by itself, identify a cause. The first obligation following a supported residual is to attack it: to search for failures of randomisation, concealment, temporal integrity, delivery, retention, and inclusion, and to require independent replication and implementation swaps.

The design generalises beyond EEG to any setting in which an outcome-relevant variable can be assigned only after a committed measurement window has closed. In a closed-loop brain-computer interface, for example, a control parameter applied after a decoding window closes plays the role of the assigned delay and the committed decoder output plays the role of the endpoint; the same boundary then asks whether the declared past-accessible information is sufficient for that output. The construction transfers wherever assignment, delivery, retention, preprocessing, and inclusion can be audited with comparable rigour, so the contribution is a reusable design-stage falsification test rather than a result.

7.1 Relation to broader theoretical interpretations

A supported and replicated residual would raise the question of what produces it. Candidate interpretations range from mundane unmodelled forward pathways to more speculative dynamical or physical accounts. All such interpretations are downstream of, and outside, the evidential claim of this article. The present test neither requires nor supports any particular mechanism, and it is designed so that its empirical conclusion stands or falls independently of any later theory. We therefore make no mechanistic claim here, and treat the construction and evaluation of explanatory models as separate, prospectively specified work to be undertaken only if and after a residual is established and replicated.

Data accessibility

This Perspective reports no human empirical data. All synthetic-benchmark numbers in the main text, Figure 2, and the electronic supplementary material are produced by a public, executable benchmark package and recorded under a run hash. The package (source code, configuration files, environment files, run scripts, tests, generated tables, figure source data, and run metadata) is available

at <https://github.com/asopasakis/cri-leveliia-benchmark> and is archived at Zenodo under DOI 10.5281/zenodo.PLACEHOLDER. The figures and tables in this article were generated from the frozen output directory with run hash 0a775ac4f1ead9d3 ($M = 1000$ Monte Carlo datasets per scenario). Reviewers can reproduce the results with `python scripts/run_all.py --all` followed by `python scripts/verify_outputs.py`.

Author contributions

Both authors developed the design and inferential framework, specified the benchmarks, and wrote the manuscript.

Conflict of interest

The authors declare no competing interests.

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